Zoo503 Subjective Questions

VU Medical Zone (Biotechnologists and Zoologists)

1.Modern Birds

Appeared in the latest Cretaceous and Early Tertiary. Flightless ratites and ancestors of water birds, penguins and birds of prey. The perching birds radiated in the Miocene.

2.Biotic Replacements

- An obvious feature of the history of life.
- These are times when one group of plants or animals replaces another.
- The replacement of brachiopods by bivalves is a famous example

3.Homo habilis

Lived in Africa from 2.4 to 1.5 Ma. Brain capacity of 630– 700 cm3 in a body only 1.3 m tall.

4.Jawless Fishes

Two key defining characters: the head and neural crest tissues

5.Redundant Species Hypothesis

Removal of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem.

6.Archaeopteryx

Archaeopteryx is the most famous fossils. Archaeopteryx is Oldest known bird found in Upper Jurassic sediments in southern Germany in 1861.

7.Traces on Bedding Planes

- **Tracks**: sets of discrete footprints, usually formed by arthropods or vertebrates
- **Trails:** continuous traces, usually formed by the whole body, either trave

8.Biogeography

Biogeography is the study of the distribution of species and ecosystems in geographic space and through geological time.

9.Geographic Information Systems

To predict future trends and understand the factors affecting organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

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10.Kleptoparasitism

A very widespread duck in the gadwall (Anas strepera). Coots (both Eurasian coot, Fulica atra, and the American coot, Fulica americana). Dive for their food and bring vegetable matter to the surface from greater depths. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called kleptoparasitism, and it is an effective way of widening the niche of the gadwall.

12.Oldest Amniote

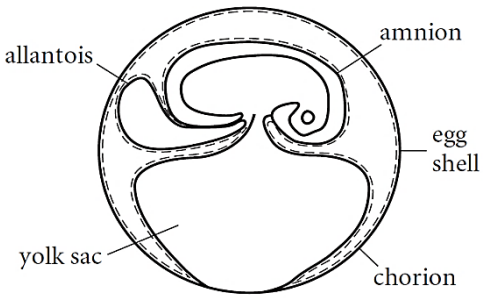
Oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada. Only skeleton; no amniotic eggs are known from the Carboniferous. E.g The earliest reptile, The mid-Carboniferous Reptile *Hylonomus*

13.Radiated amniotes

Amniotes: Late Carboniferous Three main clades distinguished by the pattern of openings in the side of the skull. Temporal openings behind the eye socket.

The three major skull patterns seen in amniotes: anapsid, diapsid and synapsid. (Based on Carroll 1987.)

- **The anapsid** ("no arch") skull pattern, no temporal openings.
- **The synapsid** ("same arch"), lower temporal opening,
- **The diapsid** ("two arch") pattern, two temporal openings.



Three Clades

- **The Anapsida**: early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles.
- **The Synapsida**: "mammal-like reptiles" and the mammals
- **The Diapsida** : Early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds. All modern amniotes produce the cleidoic egg

14.Amnionic layers of embryo of reptiles and their function

- Enclose the embryo (the amnion)
- Collect waste (the allantois)
- line the eggshell (the chorion)
- Yolk, a yellow material rich in protein.

15.Marine biogeography scientist works

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- **James Dana:** (1853), divided the surface waters into several different zones based on mean minimum temperature.
- **Edward Forbes:** (1856) published Five depth zones and 25 faunal provinces along the coasts of the continents
- In 1880, **Albert Günther** published a book on fishes in which he recognized 10 different regions in the distribution of shore fishes,
- **Arnold Ortmann** published work based on the distribution of crustaceans such as crabs and lobsters

16.Excrement

- **Fecal pellets and fecal strings**: small pellets, usually <10 mm in length, or strings of excrement
- **Coprolites**: discrete fecal masses, usually >10 mm in length, and usually the product of vertebrates

17.Niche:

The demands that an organism places on its environment in terms of physical and chemical conditions, space and food supply help to define what ecologists call its niche.Covers all aspects of the basic physics and chemistry of habitat and how the organism makes a living. Includes the food an animal requires, but also encompasses the way in which it acquires that food

18.Leon Croizat

He amassed a vast array of distributional data, representing each biogeographical pattern as a line, or track, connecting its known areas of distribution. Croizat studied the distribution patterns of many unrelated taxa, and for each he drew lines or 'tracks' on the map linking the areas in which they are found. In many cases, these lines were similar enough in position to be combined as 'generalized tracks', shown here. He found that the tracks of many taxa, belonging to a wide variety of organisms, could be combined to form a generalized track that connected different regions of the world

19.Island Biogeography of new Candolle

Pointed out that the age, climate and degree of isolation of an island, and whether or not it was volcanic, would also affect the diversity of its flora. Sheer variety and volume of the works published by Alfred Wallace

20.New Caledonia

Geological studies show that New Caledonia is the smallest (about 1600 km2) surviving fragment of Gondwana. Rifted away from Australia in the Late Cretaceous 80–65 mya and reaching its present position 1500 km east of it 50 mya. It is a hotspot for diversity.

In view of the large amount of research that shows the frequency of long-distance dispersal in such scenarios, most biogeographers today believe that this was the mechanism of arrival of the ancestors of the New Caledonian taxa. Geological studies show that the island was submerged for long periods in the Paleocene–Eocene, so that the island was not available for colonization until about 37 mya. The New Caledonian species must therefore have diverged from their ancestors after this date

- Panbiogeographers reject long-distance dispersal.They theorize that the Cretaceous ancestors of the present biota were widespread across the Pacific islands, including on supposedly formerly exposed large plateaux such as the Ontong Java plateau
- Biogeographers have investigated the history of the angiosperm family Sapotaceae. Resulted from nine colonizations of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia.

New Guinea and Australia were the most important sources of the arrivals, those from Pacific islands arriving only more recently, 27–24 mya. These results are clearly totally incompatible with the panbiogeographers' theories.

21.Old world tropic fauna of Africa

- Contains the best known & most distinctive of Old World tropical mammal fauna. Molecular studies of the interrelationships of the different orders of placentals suggest that the common ancestor of a group of African placentals entered the continent in the Late Cretaceous.
- This group is made up of the elephants, hyracoids (conies), aquatic sirenians (sea cows), elephant shrews, aardvarks, Cape golden moles, insectivorous tenrecs and extinct embrithopods; they have been placed together in a superorder, Afrotheria that is endemic to Africa.
- Much of the continent was covered by rainforests of the tropical ever wet and subtropical summer wet biomes for much of the early Cenozoic, until the Late Miocene. About 19 mya in the Miocene, a firm land connection with Eurasia took place in what is now the Middle East
- The seaway was finally broken near the beginning of the Late Miocene, 12 mya, by rising mountains in Arabia, Turkey and the Middle East. It was at this time that the early horse Hipparion, which had evolved in North America, appeared in both Eurasia and Africa

- The elephant originated in Africa in the Eocene, migrated into Eurasia in the Early Miocene and migrated back later in the Miocene The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia 2 million years later.

22.write the old history mammals of fauna of africa?

- The Old World monkeys of Africa arose from Eocene
- The apes Radiated in Africa in the Miocene.
- The apes spread out from Africa into the Middle East, Asia and southern Europe
  by the mid-Miocene
- A number of incomplete human fossils reported during the 1990s from Africa,
  dated between 6 and 4 Ma.
- Homo habilis Lived in Africa from 2.4 to 1.5 Ma
- A new species, H. erectus, which arose 1.9 Ma in Africa
- Truly modern humans, H. sapiens, may have arisen as much as 400,000 years ago,
  and certainly by 150,000 years ago, in Africa,

23.Biogeographic history of South America

- The history of the biota of South America has been dominated by the effects of the great earth-engine of plate tectonics. Westward drift of the continent, mountain building and consequent climatic changes changes in its relationship with North America.

- There had been islands in that region ever since the Americas started to move westward in the Early Cretaceous. This may have been the route by which some North American dinosaurs dispersed southward in the Mid-Cretaceous, with some early types of marsupial and placental mammal.

- But this connection has always been tenuous, sometimes permitting passage and sometimes becoming broken. This led to cycles of immigration, isolation and evolution, and final new immigration and extinction. The mammal fauna of South America today is characterized by a few marsupials (opossums) and a diversity of edentate placental mammals known as the Xenarthra (sloths, armadillos and South American anteaters) that are hardly known elsewhere.

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- In the Late Cretaceous, South America had a mammal fauna of strange marsupials and of ungulate placentals quite unlike anything known elsewhere in the world.
- Alfred Wallace, noted that the range boundaries of Amazonian birds seemed to coincide with the pattern of the rivers. It is fascinating that recent research shows that closely related taxa of some birds are to be found on either side of rivers throughout Amazonia.

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